Problem Set 4

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ILA 4.1–4.4, 6.1–6.2. Problems marked [**code**] should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016).

1. ILA, Problem Set 4.1 (orthogonality of the four subspaces): problems 1, 4.
2. ILA, Problem Set 4.2 (projections): problems 1, 4, 11.
3. ILA, Problem Set 4.4 (orthogonal matrices and Gram–Schmidt): problems 1, 5.
4. ILA, Problem Set 6.1 (introduction to eigenvalues): problems 1, 4, 9.
5. ILA, Problem Set 6.2 (diagonalization; e.g. the Fibonacci matrix): problems 1, 6.
6. [**code**] Fit a low-degree polynomial to noisy data by least-squares projection, and plot the data with the fitted curve. Separately, compute the eigenvalues of the 2×2 Fibonacci matrix and use them to explain the growth rate of the Fibonacci sequence.
7. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).